NESTED LIST AND EXAMPLE

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INTRODUCTION TO LIST

- A list is a collection of items in Python
- Lists are ordered, changeable (mutable)
- Elements can be of different data types

Example:

```
MY_LIST = [1, 2, 3, "HELLO", 4.5]
PRINT(MY_LIST)
[1, 2, 3, "HELLO", 4.5]
```

WHAT IS A NESTED LIST ?

- A nested list is a list inside another list
- It allows storing multiple lists in one list
- Used to represent matrix, tables, grids

Example:

```
NESTED_LIST = [[1, 2, 3], [4, 5, 6]]
```

STRUCTURE OF NESTED LIST

- Each element of the main list is another list
- Accessed using multiple indices

Example:

DATA = [[10, 20], [30, 40], [50, 60]]

ACCESSING ELEMENTS

- **Use indexing to access elements**

Syntax:

```
LIST[ROW][COLUMN]
```

Example:

```
DATA = [[10, 20], [30, 40]]
PRINT( DATA[0][1]  )
# OUTPUT: 20
```

APPLICATIONS OF NESTED LIST

- REPRESENTING MATRICES
- STORING TABULAR DATA
- USED IN GAMES (GRID SYSTEMS)
- USED IN DATA SCIENCE & ML

Real-Life Example

- STUDENT MARKS LIST

EXAMPLE:

```
MARKS = [
        ["MATH", 90],
        ["SCIENCE",
85],
        ["ENGLISH", 88]
        ]
```

Conclusion

- Nested lists are powerful in Python
- Help manage structured data
- Widely used in programming and real-world applications

THANK YOU